Haowen Deng

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EDUCATION

Guangdong University of Technology Guangzhou, China
Bachelor of Engineering in Software Engineering *Sept. 2023 – June 2027 Expected*

Core Coursework: Data Structures, Operating Systems, Computer Networks, Computer Organization and Architecture, Object-Oriented Programming.

EXPERIENCE

ByteDance (TikTok) - Toutiao Guangzhou, China
Android Software Engineer Intern *Dec. 2025 – July 2026*

- Refactored smart album BottomSheet with **ViewPager** nesting two-layer **RecyclerViews**; rewrote parent touch event dispatch and interception mechanisms to resolve multi-layer nested scrolling conflicts.
- Implemented album grid drag-to-reorder screen using **ItemTouchHelper**, resolving race conditions between drag displacement and item deletion animations.
- Orchestrated voice input states via a state machine; leveraged **Spannable** in **EditText** for streaming text rendering and graying. Decoupled analytics via interface callbacks for **dependency inversion** and cross-feature reuse.
- Audited and patched 10+ legacy telemetry points in publishing pipeline by filling missing fields and fixing trigger timing, supporting downstream data analytics and on-device model training.
- Audited 8-year legacy publishing pipeline to optimize API success rate; fixed workflow issues and designed execution counters across clicks, coroutines, and network request enums to diagnose anomalies and boost conversion.
- Delivered **30+ business features and fixed 60+ defects**. Patched security vulnerabilities by intercepting illicit requests via headers; resolved publishing pipeline SDK breakages caused by bytecode instrumentation on newer Android versions.
- Participated in **technical design reviews and Code Reviews** under mentor guidance, establishing defensive programming habits, rigorous engineering workflows, and team collaboration standards.

PROJECTS

Zafiro - Agent with Phone control, Shell, Python 3, with Skills and MCP support 330+ Stars · 50+ Forks · 94K LoC Kotlin
Core Developer · *GitHub* · *APK Download* *June 2026 – Present*

- Designed a unified theme token system in **Jetpack Compose** supporting Liquid Glass and Material3/3E styles, distributing component styling while scoping local **recomposition**.
- Built a permission dispatch module with priority fallback across Root, Shizuku, system dialogs, and settings, serving both standard users and elevated privilege needs.
- Engineered an Agent runtime inspired by Pi with **OkHttp** for multi-protocol abstraction, **persistent storage**, and model hot-swapping. Integrated **MCP** for workflow hooks, async tool calls, and context compression.
- Maintained 94K lines of production Kotlin with multi-module and **MVI** architecture. Enforced commit hygiene via **Git Hooks**, sustaining a Codacy A code quality rating.
- Blog: houman-dang.vercel.app/#blogs · Sources: ZafiroDialog.kt, PermissionManager.kt, Agent.kt, Okia.kt.

TECHNICAL SKILLS

- **Kotlin** and **Java** development, including core data structures, Coroutines, and Flow.
- Android development technologies including **Custom Views, Jetpack Compose, data persistence, network requests, and IPC**.
- **Solid understanding of Android core mechanisms**, including Looper mechanisms, View touch event dispatching, View rendering pipeline, and coroutine state machines.
- **Git team collaboration**, including branch management, code merging, and conflict resolution.
- Common architecture patterns including **MVP, MVVM, and MVI** for project development.
- AI-assisted development, leveraging **Git Hooks**, **MCP**, **CLI**, and **Skills** to enhance engineering quality.